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EDUCATION

BITS Pilani, Dubai Campus

B.E. Computer Science

Dubai, UAE

Expected 2027

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Machine Learning, Large Language Models, Database Management Systems

Bishop Conrad Senior Secondary School | *Senior Secondary (Class XII)*

Bareilly, India | 2023

RESEARCH & PUBLICATIONS

Lightweight Deep Learning for Tomato Leaf Disease Detection on Edge Devices

2025 – 2026

First Author; submitted to IEEE MSN 2026 (under review); Advisor: Prof. Pranav M. Pawar

- Trained a MobileNetV2 classifier (10 disease classes, PlantVillage dataset) with two-phase transfer learning, reaching **97.33% test accuracy** (macro-F1 0.963, mean AUC 0.994)
- Compressed the model to **5.3 MB / 1.4M parameters** via INT8 post-training quantization (TensorFlow Lite) with negligible accuracy loss
- Deployed the quantized model on a **Raspberry Pi 4B** for real-time on-device inference (about 250 ms/image, 3.6 FPS, 3.2 W)

EXPERIENCE

Backend Developer Intern, Amaani (Travel-Tech Startup)

Dubai, UAE

BITS Practice School (PS-1)

Jun 2025 – Aug 2025

- Developed and maintained backend services in **Python (Flask)**, building RESTful APIs for core travel-booking features
- Documented API contracts with **Swagger/OpenAPI** and validated endpoints end-to-end using **Postman**; implemented request validation and data models

FLAGSHIP PROJECTS

Agent Desk | *Next.js, FastAPI, SQLite, Claude Code*

[Live](#) | [Code](#)

- Two AI agents on one site: **Patch** audits every GitHub repository (scored out of 100) and drafts fixes; **Pitch** drafts LinkedIn posts from Patch's notes. Nothing is sent without a click: a fixed list of allowed writes, dry-run by default, a pause switch
- Claude via the Claude Code CLI (no API key), stopped at 40% plan usage; rules before model calls (12 repositories audited in 30 requests, 0 model calls); 469 pytest tests. View-only hosted copy refreshed every six hours by a GitHub Actions job

Flowboard | *TypeScript, Next.js, React, Redux Toolkit, React Flow, Zod, Docker*

[Live](#) | [Code](#)

- Visual workflow builder, a small n8n: connect eight block types, press Run, see each step's data. A React-free engine runs ready blocks in parallel; hand-written undo/redo; flows cannot run code (no **eval**, references are path lookups)
- REST relay hardened against SSRF (host allow-list, no private addresses or redirects, size and rate caps); 319 unit and 37 Playwright tests with axe scans; CI builds the Docker image and applies the Kubernetes manifest to a kind cluster

Sayso | *TypeScript, Next.js, Redux Toolkit, Zod, Tailwind CSS, Playwright*

[Live](#) | [Code](#)

- Service desk for a made-up airline: a typed or spoken request is answered with working interface (seat map, price, receipt), not text. A four-step agent (understand, plan, run, check) in plain TypeScript, 14 Zod-validated components
- The plan chooses, the code computes: the server re-prices every order and refuses a mismatch. Rules read requests with no key (76% of 105 unseen sentences before tuning); a model is optional. 370 unit and 59 end-to-end tests

Hindsight | *TypeScript, Next.js, Redux Toolkit (RTK Query), Zod, SVG*

[Live](#) | [Code](#)

- Agent observer: reads runs from the apps above through one Zod-checked trace format and draws each as a timeline (real or “agent” time), with playback and keyboard navigation. Runs arrive by **postMessage** from allow-listed origins; 289 unit, 98 end-to-end tests

Portfolio OS | *TypeScript, Next.js, React, Redux Toolkit, Zod, Playwright*

[Live](#) | [Code](#)

- Personal portfolio built as a Windows-style desktop: projects are icons that open in windows, and the live apps above run inside theirs. Window manager written from scratch (drag, resize, snap, keyboard), with unit-tested math
- An assistant answers questions about the owner from a fact base, with sources, in plain TypeScript, no API or model; first-run accuracy 83% on 166 questions written before tuning, 67.5% on 40 held-out. 249 unit and 131 end-to-end tests with axe

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, HTML/CSS, Python, Java, SQL, C, LaTeX

Front-End: React, Next.js, state management (Redux Toolkit, Context), React Flow, Tailwind CSS, CSS animation, responsive design, data visualization (SVG charts), interactive UI, Zod, accessibility (WCAG, axe)

Generative AI / LLMs: Claude API & Claude Code, Gemini API, agentic tool-calling, prompt engineering, RAG, FAISS, structured output (Zod, Pydantic)

Machine Learning: NumPy, Pandas, scikit-learn, TensorFlow/Keras, TensorFlow Lite, transfer learning, edge ML (Raspberry Pi)

Backend & Databases: Node.js, FastAPI, Flask, REST API design, Swagger/OpenAPI, Postman, SQLite, MySQL

Testing & DevOps: Vitest, Playwright, pytest, CI/CD (GitHub Actions), Docker, Kubernetes, Vercel, Git, AWS (learning)

CERTIFICATIONS

The Ultimate Job Ready Data Science Course – CodeWithHarry (2026) · AWS Cloud Technical Essentials – Coursera (*in progress*)